

OSSP Practical 10:

Practical 10 — Theory

Linux File System Internals and VFS

Linux uses a **file system** to organize and manage files and directories on storage devices. Each file is associated with an **inode**, which stores information such as file permissions, owner, size, timestamps, and file type.

The **Virtual File System (VFS)** is a layer in the Linux kernel that provides a common interface for different file systems. It allows applications to access files without needing to know which specific file-system implementation is being used.

Important Concepts

- **File System:** Organizes and stores files and directories.
- **Inode:** Contains metadata about a file, such as permissions, owner, size, and timestamps.
- **Directory:** Stores references to files and their inode numbers.
- **Mount Point:** A directory where a file system is attached and made accessible.
- **VFS:** Provides a common interface between applications and different file systems.
- **/proc/filesystems:** Shows file-system types supported by the Linux kernel.
- **/proc/mounts:** Shows currently mounted file systems.
- **stat:** Displays detailed information about a file or directory.
- **df:** Displays disk-space usage and file-system information.
- **findmnt:** Displays mounted file-system information.

Conclusion

Linux file-system internals manage files, directories, metadata, and storage. The VFS provides a standard interface that allows Linux to work with different file-system types efficiently.

```

alurusaranya@LAPTOP-0PA61QEN:~$ mkdir practical10
alurusaranya@LAPTOP-0PA61QEN:~$ cd practical10
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ pwd
/home/alurusaranya/practical10
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ df -T
Filesystem      Type      1K-blocks      Used Available Use% Mounted on
none            overlay   3948604         0    3948604    0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            tmpfs     3948604         4    3948600    1% /mnt/wsl
drivers         9p        497581052      156427716 341153336 32% /usr/lib/wsl/drivers
/dev/sdd        ext4      1055762868     2544404 999514992 1% /
none            tmpfs     3948604         0    3948564    1% /mnt/wslg
none            overlay   3948604         0    3948604    0% /usr/lib/wsl/lib
rootfs          rootfs    3942848        2772    3940076    1% /init
none            tmpfs     3948604         0    3948084    1% /run
none            tmpfs     3948604         0    3948604    0% /run/lock
none            tmpfs     3948604         0    3948604    0% /run/shm
none            overlay   3948604         80    3948524    1% /mnt/wslg/versions.txt
none            overlay   3948604         80    3948524    1% /mnt/wslg/doc
C:\             9p        497581052      156427716 341153336 32% /mnt/c
tmpfs           tmpfs     789720          20    789700    1% /run/user/0
tmpfs           tmpfs     789720          20    789700    1% /run/user/1000
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ cat /proc/filesystems
nodev          sysfs
nodev          tmpfs
nodev          bdev
nodev          proc
nodev          cgroup
nodev          cgroup2
nodev          cpuset
nodev          devtmpfs
nodev          binfmt_misc
nodev          configfs
nodev          debugfs
nodev          tracefs
nodev          securityfs
nodev          sockfs
nodev          bpf

nodev          sockfs
nodev          bpf
nodev          pipefs
nodev          ramfs
nodev          hugetlbfs
nodev          devpts
nodev          ext3
nodev          ext4
nodev          ext2
nodev          squashfs
nodev          vfat
nodev          fuseblk
nodev          fuse
nodev          fusectl
nodev          virtiofs
nodev          overlay
nodev          udf
nodev          xfs
nodev          9p
nodev          efivarfs
nodev          mqueue
nodev          selinuxfs
nodev          pstore
nodev          autofs
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ ls -l /proc/filesystems
-r--r--r-- 1 root root 0 Sep 28 04:46 /proc/filesystems
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ cat /proc/mounts
none /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 overlay rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uuid=on 0 0
none /mnt/wsl tmpfs rw,relatime 0 0
drivers /usr/lib/wsl/drivers 9p ro,nosuid,nodev,noatime,aname=drivers;fmask=222;dmask=222,cache=0x5,access=client,msize=65536,trans=f
d,rfd=8,wfd=8 0 0
/dev/sdd / ext4 rw,relatime,discard,errors=remount-ro,data=ordered 0 0
none /mnt/wslg tmpfs rw,relatime 0 0
/dev/sdd /mnt/wslg/distro ext4 ro,relatime,discard,errors=remount-ro,data=ordered 0 0
none /usr/lib/wsl/lib overlay rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/g
pu_lib/rw/work,uuid=on 0 0

```

```

pu_lib/rw/work,uid=on 0 0
rootfs /init rootfs ro,size=3942848k,nr_inodes=985712 0 0
none /dev devtmpfs rw,nosuid,relatime,size=3942848k,nr_inodes=985712,mode=755 0 0
sysfs /sys sysfs rw,nosuid,nodev,noexec,noatime 0 0
proc /proc proc rw,nosuid,nodev,noexec,noatime 0 0
devpts /dev/pts devpts rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000 0 0
none /run tmpfs rw,nosuid,nodev,mode=755 0 0
none /run/lock tmpfs rw,nosuid,nodev,noexec,noatime 0 0
none /run/shm tmpfs rw,nosuid,nodev,noatime 0 0
none /dev/shm tmpfs rw,nosuid,nodev,noatime 0 0
none /run/user tmpfs rw,nosuid,nodev,noexec,noatime,mode=755 0 0
binfmt_misc /proc/sys/fs/binfmt_misc binfmt_misc rw,relatime 0 0
cgroup2 /sys/fs/cgroup cgroup2 rw,nosuid,nodev,noexec,relatime,nsdelegate 0 0
none /mnt/wslg/versions.txt overlay rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on 0 0
none /mnt/wslg/doc overlay rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on 0 0
none /tmp.X11-unix tmpfs ro,relatime 0 0
C:\134 /mnt/c 9p rw,noatime,aname=drvfs,path=C:\;uid=0;gid=0;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6 0 0
none /run/user tmpfs rw,relatime 0 0
hugetlbfs /dev/hugepages hugetlbfs rw,nosuid,nodev,relatime,pagesize=2M 0 0
mqueue /dev/mqueue mqueue rw,nosuid,nodev,noexec,relatime 0 0
debugfs /sys/kernel/debug debugfs rw,nosuid,nodev,noexec,relatime 0 0
tracefs /sys/kernel/tracing tracefs rw,nosuid,nodev,noexec,relatime 0 0
fusectl /sys/fs/fuse/connections fusectl rw,nosuid,nodev,noexec,relatime 0 0
configfs /sys/kernel/config configfs rw,nosuid,nodev,noexec,relatime 0 0
tmpfs /run/user/0 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700 0 0
tmpfs /mnt/wslg/run/user/0 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700 0 0
tmpfs /run/user/1000 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700,uid=1000,gid=1000 0 0
tmpfs /mnt/wslg/run/user/1000 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700,uid=1000,gid=1000 0 0
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ mount | head -20
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uid=on)
none on /mnt/wsl type tmpfs (rw,relatime)
drivers on /usr/lib/wsl/drivers type 9p (ro,nosuid,nodev,noatime,aname=drivers,fmask=222,dmask=222,cache=0x5,access=client,msize=65536,trans=fd,rfd=8,wfd=8)
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /mnt/wslg type tmpfs (rw,relatime)

6,trans=fd,rfd=8,wfd=8)
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /mnt/wslg type tmpfs (rw,relatime)
/dev/sdd on /mnt/wslg/distro type ext4 (ro,relatime,discard,errors=remount-ro,data=ordered)
none on /usr/lib/wsl/lib type overlay (rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work,uid=on)
rootfs on /init type rootfs (ro,size=3942848k,nr_inodes=985712)
none on /dev type devtmpfs (rw,nosuid,relatime,size=3942848k,nr_inodes=985712,mode=755)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000)
none on /run type tmpfs (rw,nosuid,nodev,mode=755)
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /dev/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)
none on /mnt/wslg/versions.txt type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on)
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ df -hT

```

Filesystem	Type	Size	Used	Avail	Use%	Mounted on
none	overlay	3.8G	0	3.8G	0%	/usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none	tmpfs	3.8G	4.0K	3.8G	1%	/mnt/wsl
drivers	9p	475G	150G	326G	32%	/usr/lib/wsl/drivers
/dev/sdd	ext4	1007G	2.5G	954G	1%	/
none	tmpfs	3.8G	40K	3.8G	1%	/mnt/wslg
none	overlay	3.8G	0	3.8G	0%	/usr/lib/wsl/lib
rootfs	rootfs	3.8G	2.8M	3.8G	1%	/init
none	tmpfs	3.8G	520K	3.8G	1%	/run
none	tmpfs	3.8G	0	3.8G	0%	/run/lock
none	tmpfs	3.8G	0	3.8G	0%	/run/shm
none	overlay	3.8G	80K	3.8G	1%	/mnt/wslg/versions.txt
none	overlay	3.8G	80K	3.8G	1%	/mnt/wslg/doc
C:\	9p	475G	150G	326G	32%	/mnt/c
tmpfs	tmpfs	772M	20K	772M	1%	/run/user/0
tmpfs	tmpfs	772M	20K	772M	1%	/run/user/1000

```

total 8
drwxrwxr-x  2 alurusaranya alurusaranya 4096 Sep 28 05:06 .
drwxr-x--- 13 alurusaranya alurusaranya 4096 Sep 28 05:06 ..
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ ls -li
total 0
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ stat .
  File: .
  Size: 4096          Blocks: 8          IO Block: 4096   directory
Device: 8,48    Inode: 12165          Links: 2
Access: (0775/drwxrwxr-x)  Uid: ( 1000/alurusaranya)   Gid: ( 1000/alurusaranya)
Access: 2026-09-28 05:08:45.612208264 +0000
Modify: 2026-09-28 05:06:24.876385251 +0000
Change: 2026-09-28 05:06:24.876385251 +0000
 Birth: 2026-09-28 05:06:24.876385251 +0000
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ touch test.txt
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ echo "Linux File System Internals" > test.txt
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ ls -li test.txt
12170 -rw-rw-r--  1 alurusaranya alurusaranya 28 Sep 28 05:09 test.txt
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ stat test.txt
  File: test.txt
  Size: 28          Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 12170          Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/alurusaranya)   Gid: ( 1000/alurusaranya)
Access: 2026-09-28 05:09:09.207202266 +0000
Modify: 2026-09-28 05:09:13.731867767 +0000
Change: 2026-09-28 05:09:13.731867767 +0000
 Birth: 2026-09-28 05:09:09.207202266 +0000
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ cat test.txt
Linux File System Internals
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ findmnt .
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ df -hT .
Filesystem      Type      Size  Used Avail Use% Mounted on
/dev/sdd         ext4    1007G  2.5G  954G   1% /
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ mount | head -20
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uuid=on)
none on /mnt/wsl type tmpfs (rw,relatime)
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ mount | head -20
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uuid=on)
none on /mnt/wsl type tmpfs (rw,relatime)
drivers on /usr/lib/wsl/drivers type 9p (ro,nosuid,nodev,noatime,aname=drivers,fmask=222,dmask=222,cache=0x5,access=client,msize=65536,trans=fd,rfd=8,wfd=8)
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /mnt/wslg type tmpfs (rw,relatime)
/dev/sdd on /mnt/wslg/distro type ext4 (ro,relatime,discard,errors=remount-ro,data=ordered)
none on /usr/lib/wsl/lib type overlay (rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work,uuid=on)
rootfs on /init type rootfs (ro,size=3942848k,nr_inodes=985712)
none on /dev type devtmpfs (rw,nosuid,relatime,size=3942848k,nr_inodes=985712,mode=755)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000)
none on /run type tmpfs (rw,nosuid,nodev,mode=755)
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /dev/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)
none on /mnt/wslg/versions.txt type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uuid=on)
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ cat /proc/filesystems
nodev      sysfs
nodev      tmpfs
nodev      bdev
nodev      proc
nodev      cgroup
nodev      cgroup2
nodev      cpuset
nodev      devtmpfs
nodev      binfmt_misc
nodev      configfs

```

```

nodev    cgroup2
nodev    cpuset
nodev    devtmpfs
nodev    binfmt_misc
nodev    configfs
nodev    debugfs
nodev    tracefs
nodev    securityfs
nodev    sockfs
nodev    bpf
nodev    pipefs
nodev    ramfs
nodev    hugetlbfs
nodev    devpts
nodev    ext3
nodev    ext4
nodev    ext2
nodev    squashfs
nodev    vfat
nodev    fuseblk
nodev    fuse
nodev    fusectl
nodev    virtiofs
nodev    overlay
nodev    udf
nodev    xfs
nodev    9p
nodev    efivarfs
nodev    mqueue
nodev    selinuxfs
nodev    pstore
nodev    autofs
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ cat /proc/mounts | grep -E "ext4|tmpfs"
none /mnt/wsl tmpfs rw,relatime 0 0
/dev/sdd / ext4 rw,relatime,discard,errors=remount-ro,data=ordered 0 0
none /mnt/wslg tmpfs rw,relatime 0 0
/dev/sdd /mnt/wslg/distro ext4 ro,relatime,discard,errors=remount-ro,data=ordered 0 0

/dev/sdd /mnt/wslg/distro ext4 ro,relatime,discard,errors=remount-ro,data=ordered 0 0
none /dev devtmpfs rw,nosuid,relatime,size=3942848k,nr_inodes=985712,mode=755 0 0
none /run tmpfs rw,nosuid,nodev,mode=755 0 0
none /run/lock tmpfs rw,nosuid,nodev,noexec,noatime 0 0
none /run/shm tmpfs rw,nosuid,nodev,noatime 0 0
none /dev/shm tmpfs rw,nosuid,nodev,noatime 0 0
none /run/user tmpfs rw,nosuid,nodev,noexec,noatime,mode=755 0 0
none /tmp/.X11-unix tmpfs ro,relatime 0 0
none /run/user tmpfs rw,relatime 0 0
tmpfs /run/user/0 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700 0 0
tmpfs /mnt/wslg/run/user/0 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700 0 0
tmpfs /run/user/1000 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700,uid=1000,gid=1000 0 0
tmpfs /mnt/wslg/run/user/1000 tmpfs rw,nosuid,nodev,relatime,size=789720k,nr_inodes=197430,mode=700,uid=1000,gid=1000 0 0
alurusaranya@LAPTOP-0PA61QEN:~/practical10$ ls -l /dev
total 0
crwxr-xr-x 1 root root    10, 235 Sep 28 04:46 autofs
drwxr-xr-x 2 root root   600 Sep 28 04:46 block
drwxr-xr-x 2 root root   120 Sep 28 04:46 bsg
crw----- 1 root root    10, 234 Sep 28 04:46 btrfs-control
drwxr-xr-x 2 root root  2860 Sep 28 04:46 char
crw----- 1 root root     5,   1 Sep 28 04:46 console
lrwxrwxrwx 1 root root    11 Sep 28 04:46 core -> /proc/kcore
crw----- 1 root root   10, 260 Sep 28 04:46 cpu_dma_latency
crw----- 1 root root   10, 203 Sep 28 04:46 cuse
drwxr-xr-x 6 root root   120 Sep 28 04:46 disk
crw-rw-rw- 1 root root   10, 258 Sep 28 04:46 dxg
lrwxrwxrwx 1 root root    13 Sep 28 04:46 fd -> /proc/self/fd
crw-rw-rw- 1 root root     1,   7 Sep 28 04:46 full
crw-rw-rw- 1 root root   10, 229 Sep 28 04:46 fuse
crw----- 1 root root   10, 228 Sep 28 04:46 hpet
drwxr-xr-x 2 root root     0 Sep 28 04:46 hugepages
crw----- 1 root root  229,   0 Sep 28 04:46 hvc0
crw----- 1 root root  229,   1 Sep 28 05:04 hvc1
crw--w---- 1 root tty   229,   2 Sep 28 04:46 hvc2
crw----- 1 root root  229,   3 Sep 28 04:46 hvc3
crw----- 1 root root  229,   4 Sep 28 04:46 hvc4
crw----- 1 root root  229,   5 Sep 28 04:46 hvc5

```

2520080012

```
crw----- 1 root root 229, 1 Sep 28 05:04 hvc1
crw--w---- 1 root tty 229, 2 Sep 28 04:46 hvc2
crw----- 1 root root 229, 3 Sep 28 04:46 hvc3
crw----- 1 root root 229, 4 Sep 28 04:46 hvc4
crw----- 1 root root 229, 5 Sep 28 04:46 hvc5
crw----- 1 root root 229, 6 Sep 28 04:46 hvc6
crw----- 1 root root 229, 7 Sep 28 04:46 hvc7
crw----- 1 root root 10, 183 Sep 28 04:46 hwrng
lrwxrwxrwx 1 root root 12 Sep 28 04:46 initctl -> /run/initctl
crw-r--r-- 1 root root 1, 11 Sep 28 04:46 kmsg
crw-rw---- 1 root kvm 10, 232 Sep 28 04:46 kvm
lrwxrwxrwx 1 root root 28 Sep 28 04:46 log -> /run/systemd/journal/dev-log
crw-rw---- 1 root disk 10, 237 Sep 28 04:46 loop-control
brw-rw---- 1 root disk 7, 0 Sep 28 04:46 loop0
brw-rw---- 1 root disk 7, 1 Sep 28 04:46 loop1
brw-rw---- 1 root disk 7, 2 Sep 28 04:46 loop2
brw-rw---- 1 root disk 7, 3 Sep 28 04:46 loop3
brw-rw---- 1 root disk 7, 4 Sep 28 04:46 loop4
brw-rw---- 1 root disk 7, 5 Sep 28 04:46 loop5
brw-rw---- 1 root disk 7, 6 Sep 28 04:46 loop6
brw-rw---- 1 root disk 7, 7 Sep 28 04:46 loop7
drwxr-xr-x 2 root root 60 Sep 28 04:46 mapper
crw----- 1 root root 10, 227 Sep 28 04:46 mcelog
crw-r----- 1 root kmem 1, 1 Sep 28 04:46 mem
crw----- 1 root root 10, 220 Sep 28 04:46 mptctl
drwxrwxrwt 2 root root 40 Sep 28 04:46 mqueue
drwxr-xr-x 2 root root 60 Sep 28 04:46 net
crw-rw-rw- 1 root root 1, 3 Sep 28 04:46 null
crw----- 1 root root 10, 144 Sep 28 04:46 nvram
crw----- 1 root root 108, 0 Sep 28 04:46 ppp
crw-rw-rw- 1 root tty 5, 2 Sep 28 05:10 ptmx
crw----- 1 root root 246, 0 Sep 28 04:46 ptp0
lrwxrwxrwx 1 root root 4 Sep 28 04:46 ptp_hyperv -> ptp0
drwxr-xr-x 2 root root 0 Sep 28 04:46 pts
brw-rw---- 1 root disk 1, 0 Sep 28 04:46 ram0
brw-rw---- 1 root disk 1, 1 Sep 28 04:46 ram1
brw-rw---- 1 root disk 1, 10 Sep 28 04:46 ram10
```

```
drwxr-xr-x 2 root root 60 Sep 28 04:46 snd
lrwxrwxrwx 1 root root 15 Sep 28 04:46 stderr -> /proc/self/fd/2
lrwxrwxrwx 1 root root 15 Sep 28 04:46 stdin -> /proc/self/fd/0
lrwxrwxrwx 1 root root 15 Sep 28 04:46 stdout -> /proc/self/fd/1
crw-rw-rw- 1 root tty 5, 0 Sep 28 04:46 tty
crw--w---- 1 root tty 4, 0 Sep 28 04:46 tty0
crw--w---- 1 root tty 4, 1 Sep 28 04:46 tty1
crw--w---- 1 root tty 4, 10 Sep 28 04:46 tty10
crw--w---- 1 root tty 4, 11 Sep 28 04:46 tty11
crw--w---- 1 root tty 4, 12 Sep 28 04:46 tty12
crw--w---- 1 root tty 4, 13 Sep 28 04:46 tty13
crw--w---- 1 root tty 4, 14 Sep 28 04:46 tty14
crw--w---- 1 root tty 4, 15 Sep 28 04:46 tty15
crw--w---- 1 root tty 4, 16 Sep 28 04:46 tty16
crw--w---- 1 root tty 4, 17 Sep 28 04:46 tty17
crw--w---- 1 root tty 4, 18 Sep 28 04:46 tty18
crw--w---- 1 root tty 4, 19 Sep 28 04:46 tty19
crw--w---- 1 root tty 4, 2 Sep 28 04:46 tty2
crw--w---- 1 root tty 4, 20 Sep 28 04:46 tty20
crw--w---- 1 root tty 4, 21 Sep 28 04:46 tty21
crw--w---- 1 root tty 4, 22 Sep 28 04:46 tty22
crw--w---- 1 root tty 4, 23 Sep 28 04:46 tty23
crw--w---- 1 root tty 4, 24 Sep 28 04:46 tty24
crw--w---- 1 root tty 4, 25 Sep 28 04:46 tty25
crw--w---- 1 root tty 4, 26 Sep 28 04:46 tty26
crw--w---- 1 root tty 4, 27 Sep 28 04:46 tty27
crw--w---- 1 root tty 4, 28 Sep 28 04:46 tty28
crw--w---- 1 root tty 4, 29 Sep 28 04:46 tty29
crw--w---- 1 root tty 4, 3 Sep 28 04:46 tty3
crw--w---- 1 root tty 4, 30 Sep 28 04:46 tty30
crw--w---- 1 root tty 4, 31 Sep 28 04:46 tty31
crw--w---- 1 root tty 4, 32 Sep 28 04:46 tty32
crw--w---- 1 root tty 4, 33 Sep 28 04:46 tty33
crw--w---- 1 root tty 4, 34 Sep 28 04:46 tty34
crw--w---- 1 root tty 4, 35 Sep 28 04:46 tty35
crw--w---- 1 root tty 4, 36 Sep 28 04:46 tty36
crw--w---- 1 root tty 4, 37 Sep 28 04:46 tty37
```

2520080012

```
crw-rw---- 1 root dialout 4, 70 Sep 28 04:46 ttyS6
crw-rw---- 1 root dialout 4, 71 Sep 28 04:46 ttyS7
crw-rw-rw-rw- 1 root root 10, 223 Sep 28 04:46 uinput
crw-rw-rw-rw- 1 root root 1, 9 Sep 28 04:46 urandom
crw-rw-rw-rw- 1 root root 10, 257 Sep 28 04:46 userfaultfd
crw-rw---- 1 root tty 7, 0 Sep 28 04:46 vcs
crw-rw---- 1 root tty 7, 1 Sep 28 04:46 vcs1
crw-rw---- 1 root tty 7, 2 Sep 28 04:46 vcs2
crw-rw---- 1 root tty 7, 3 Sep 28 04:46 vcs3
crw-rw---- 1 root tty 7, 4 Sep 28 04:46 vcs4
crw-rw---- 1 root tty 7, 5 Sep 28 04:46 vcs5
crw-rw---- 1 root tty 7, 6 Sep 28 04:46 vcs6
crw-rw---- 1 root tty 7, 128 Sep 28 04:46 vcsa
crw-rw---- 1 root tty 7, 129 Sep 28 04:46 vcsa1
crw-rw---- 1 root tty 7, 130 Sep 28 04:46 vcsa2
crw-rw---- 1 root tty 7, 131 Sep 28 04:46 vcsa3
crw-rw---- 1 root tty 7, 132 Sep 28 04:46 vcsa4
crw-rw---- 1 root tty 7, 133 Sep 28 04:46 vcsa5
crw-rw---- 1 root tty 7, 134 Sep 28 04:46 vcsa6
crw-rw---- 1 root tty 7, 64 Sep 28 04:46 vcsu
crw-rw---- 1 root tty 7, 65 Sep 28 04:46 vcsu1
crw-rw---- 1 root tty 7, 66 Sep 28 04:46 vcsu2
crw-rw---- 1 root tty 7, 67 Sep 28 04:46 vcsu3
crw-rw---- 1 root tty 7, 68 Sep 28 04:46 vcsu4
crw-rw---- 1 root tty 7, 69 Sep 28 04:46 vcsu5
crw-rw---- 1 root tty 7, 70 Sep 28 04:46 vcsu6
drwxr-xr-x 2 root root 60 Sep 28 04:46 vfio
crw-rw-rw-rw- 1 root root 10, 256 Sep 28 04:46 vga_arbiter
crw-rw-rw-rw- 1 root kvm 10, 238 Sep 28 04:46 vhost-net
crw-rw-rw-rw- 1 root kvm 10, 241 Sep 28 04:46 vhost-vsock
drwxr-xr-x 2 root root 100 Sep 28 04:46 virtio-ports
crw-rw-rw-rw- 1 root root 243, 0 Sep 28 04:46 vport0p0
crw-rw-rw-rw- 1 root root 243, 1 Sep 28 04:46 vport0p1
crw-rw-rw-rw- 1 root root 243, 2 Sep 28 04:46 vport0p2
crw-rw-rw-rw- 1 root root 10, 259 Sep 28 04:46 vsock
crw-rw-rw-rw- 1 root root 1, 5 Sep 28 04:46 zero
alurusaranya@LAPTOP-0PA61QEN:~/practical10$
```